

Employment

Product Engineer Intern	Isomer AI	May 2025 – May 2026
- Shipped 10+ production AI features powering real user workflows: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic orchestration, MCP tooling, and real-time event processing; 923 commits, 7 production deployments. - Own PostgreSQL, Redis, Docker, Sentry; build document pipelines and resolve production incidents.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deploy cloud genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets. - Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%. - Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

AlgoPulse	Agentic AI Platform	2025
- Built end-to-end agentic AI platform delivering real-time reasoning streams to React Native mobile clients via Edge Functions; eliminated API-polling latency bottleneck; Temporal.io durable workflows; 2,000 req/sec sustained. - Kafka + Temporal.io for fault-tolerant state sync; zero message loss at 2,000 concurrent AI inference requests per second.		
LLM Eval Harness	Independent Project	2025
- Built Python evaluation framework measuring quality, reliability, and performance of Claude, GPT-4o, and Gemini across 2,000 labeled examples spanning factual recall, multi-step reasoning, and code generation to inform production model selection. - Tracked hallucination rate, p50/p95 latency, cost-per-call; bootstrap CIs + McNemar tests across 6 metrics.		
KubeServe	Kubernetes LLM Inference Platform	2026
- Optimized LLM inference for resource efficiency: KEDA autoscaler scales pods 0→N on queue depth and to zero when idle (eliminating idle compute cost); FastAPI router with circuit breaker and EWMA latency-aware load balancing across providers. - Instrumented OTel + Prometheus + Grafana observability; Helm + ArgoCD GitOps; k6 to 500 VUs; 88 pytest tests.		
PerCent	GenAI Finance App – HackNC24 Top 10	2024
- Built consumer-facing React Native GenAI finance app for iOS and Android: RAG pipeline (Hugging Face + Qwen-14B-Chat) powering personalized AI financial insights; goal tracking and expense logging for end users; ranked Top 10 of 100+ teams. - RAG pipeline: Selenium scraping + Hugging Face embeddings + Qwen-14B-Chat; reduced manual research by 30% per user.		

Languages and Technologies

- Languages:** Python, Java, TypeScript, JavaScript, SQL, Bash
- Mobile & AI:** React Native, Android (Java), Claude/GPT-4o/Gemini APIs, LangChain, MCP, RAG, Hugging Face
- Performance & Systems:** Kubernetes, KEDA, FastAPI, Temporal.io, Kafka, Docker, PostgreSQL, Redis, AWS, GCP
- Quality & Observability:** OpenTelemetry, Prometheus, Grafana, Testcontainers, pytest, k6, GitHub Actions

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog